SukariSol

An Integrated Digital Intelligence Platform for Kenya’s Sugar Value Chain

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Abstract

Kenya’s sugar value chain depends on coordinated decisions across farms, harvesting, transport, milling, farmer payments and product distribution. Yet relevant information is often held in disconnected records, limiting visibility and timely action. SukariSol is a proposed integrated digital platform that connects six interoperable modules: **CaneYield** for geospatial farm and crop intelligence; **CaneFlow** for harvest and transport coordination; **FairCane** for delivery and payment reconciliation; **MillMind** for factory maintenance intelligence; **ByProduct Hub** for circular-economy opportunities; and **SugarMarket** for demand, inventory and distribution planning.

The platform will combine farm records, satellite-derived observations, logistics data and factory information to support evidence-informed decisions. A phased implementation will begin with stakeholder research, data validation and a focused prototype before progressing to pilot testing with participating industry partners. Expected benefits include improved operational visibility, more traceable transactions, better supply planning and identification of value-addition opportunities. These benefits will be measured during pilots rather than assumed in advance.

Keywords: digital agriculture; sugarcane; geospatial intelligence; logistics optimization; factory efficiency; value-chain integration

1. Introduction

Kenya’s sugar industry brings together smallholder farmers, out-grower institutions, transporters, sugar mills, distributors and regulators. Decisions in one part of the chain affect others: farm location and crop condition influence harvest planning; harvest schedules affect truck allocation and factory intake; delivery records inform farmer accounts; and mill output and inventory influence distribution. When information is fragmented or delayed, coordinating these activities becomes more difficult.

The Kenya Sugar Board’s 2026 Innovation Call emphasizes practical innovations with a credible pathway to industry application. It identifies climate-smart agronomy, regional competitiveness, financing and investment, out-grower institutions, consumer protection and market diversification, alongside AI and industrial automation, as relevant areas for innovation [1]. National agriculture reporting has also identified operational constraints in the sector, including harvesting and collection delays, payment issues and factory performance challenges [2].

SukariSol proposes one connected platform rather than a set of isolated applications. Its purpose is to help industry participants access authorized, relevant information and act on it through linked workflows. The concept remains subject to stakeholder validation, data-access agreements and technical testing; the proposed benefits are targets for evaluation, not proven results.

2. Proposed Innovation

SukariSol uses a modular architecture in which each module can provide standalone value while sharing approved records through common services, interfaces and access controls.

CaneYield

Geospatial farm and crop intelligence

Registers farm boundaries and acreage; maps farm locations and road distances to mills; displays available satellite observations and vegetation indicators; and combines these with planting dates, crop stage and historical harvest records to support field monitoring and transparent production estimates. Satellite indicators are treated as proxies for vegetation condition, not direct measurements of tonnes or definitive proof of crop type.

CaneFlow

Harvest and logistics optimization

Links eligible farms, collection requests, truck capacity, road routes and mill receiving slots. It recommends dispatch schedules and tracks collection and delivery events. Initial optimization uses transparent rules and constraints; performance is compared with a defined baseline.

FairCane

Delivery and payment reconciliation

Connects delivery references and authorized weighbridge records with applicable pricing rules, itemized deductions, payment status and dispute workflows. Discrepancies trigger review, not automatic accusations or determinations of fault.

MillMind

Factory reliability intelligence

Maintains equipment registers, fault and repair histories, downtime records, inspection schedules and spare-parts information. Rule-based alerts and maintenance analytics are developed first. Predictive models will be considered only if sufficiently complete, representative historical or sensor data becomes available.

ByProduct Hub

Circular-economy commercialization

Records available bagasse, molasses, filter cake and other eligible materials, subject to existing uses and contracts. It matches verified listings with potential buyers and compares indicative revenue with quality, handling, storage and transport costs. It does not assume all by-products are surplus or available for sale.

SugarMarket

Demand and distribution intelligence

Uses authorized sales, order, stock and transport records to show inventory positions, flag potential stock imbalances and compare replenishment scenarios. Forecasts are validated against historical data before operational use.

Shared layer. A SukariSol Command Center provides role-specific dashboards and cross-module indicators. Shared platform services include identity and organization management, permissions, audit trails, notifications, reporting, data import/export and documented interfaces. A modular monolith is proposed for the initial build to reduce delivery complexity, with selective service separation only when justified by workload or integration needs.

3. Methodology and Implementation

The project follows a five-month research-and-development pathway.

Phase	Focus and activities
Phase 1 Month 1	**Stakeholder and process research** Interview farmers, out-grower organizations, transporters, agronomists, mill operations and maintenance teams, finance staff and distributors. Map current processes and document pain points, existing software, decision-makers and willingness to participate in a pilot.
Phase 2 Month 2	**Data and technical feasibility** Inventory available farm, harvest, weighbridge, payment, maintenance, sales and inventory data. Test geospatial mapping and satellite imagery over selected sugarcane-growing areas. Assess cloud cover, farm size, road-data quality, data rights and integration constraints. Review existing services, including Miwa Bora and other relevant systems, to establish differentiation rather than duplicate capabilities.
Phase 3 Month 3	**Validation and specification** Test clickable prototypes with intended users; prioritize features using problem severity, customer value, data readiness, feasibility, differentiation and potential impact. Agree on pilot scope, access permissions, baseline metrics and evaluation procedures with prospective partners.
Phase 4 Month 4	**Integrated prototype** Build shared identity and farm records, a geospatial interface, basic satellite observations, harvest scheduling, delivery records and payment reconciliation. Add initial maintenance and commercial dashboards where data is available.
Phase 5 Month 5	**Integration and pilot readiness** Test role-based access, calculations, audit trails, mobile usability, data quality and workflow integration. Prepare documentation, training materials and a pilot report template.

First demonstrator. The first demonstrator focuses on one connected workflow: register and map a farm; record crop stage and an estimated harvest window; create a collection order; assign a truck; record delivery; reconcile the delivery against payment records; and display the resulting status to authorized users. Other modules are presented as progressively implemented components, with clear labels distinguishing functional features, simulations and planned capabilities.

4. Expected Value and Evaluation

SukariSol aims to improve coordination and information quality across the value chain. Potential value includes more visible farm and crop information, better-planned cane collection, reduced time spent tracing records, clearer delivery-to-payment reconciliation, improved maintenance planning, and new opportunities to commercialize suitable by-products. The platform may also support more informed distribution and inventory decisions.

Impact will be assessed using baseline and pilot measurements rather than unsupported savings claims. Proposed indicators by module:

CaneYield	Proportion of registered farms with validated boundaries and complete records; difference between estimated and observed harvest volumes where reliable ground data exists.
CaneFlow	Truck waiting time, empty travel and delivery punctuality.
FairCane	Proportion of deliveries successfully reconciled; time to resolve discrepancies.
MillMind	Equipment downtime and maintenance response times.
ByProduct Hub	By-product listings converted into qualified enquiries or transactions.
SugarMarket	Forecast error; stockout and excess-stock indicators.

Risks and mitigations. The main risks are incomplete or inconsistent data, limited integration with existing systems, variable satellite coverage, inaccurate yield estimates, low user adoption, connectivity constraints and concerns over commercially or personally sensitive information. Mitigations include field validation, transparent uncertainty labels, offline-friendly workflows where feasible, user training, consent and role-based access, audit logs, and staged integration. Farmer and factory records will not be made universally visible; each role will access only the information needed for its authorized tasks.

5. Commercialization Pathway

Customers and revenue. The initial target customers are sugar mills and out-grower institutions, with transport contractors, distributors and selected service providers as additional users. A mill-funded subscription or enterprise licence could support shared use across its grower and logistics network. Implementation, data migration, staff training and integration could be priced separately. Optional advanced geospatial analytics, forecasting and maintenance analytics may be offered once their value is validated. The business model and willingness to pay will be tested during stakeholder research.

Pilot. The first pilot should involve one willing mill or out-grower organization and a defined group of farms and transport workflows. It should establish a baseline, use agreed data-sharing arrangements, and test whether the system improves visibility or measurable operational outcomes without disrupting existing operations. Expansion should depend on pilot results, user feedback, data quality and a viable support model.

Differentiation. SukariSol's differentiation is the integration of farm intelligence, logistics, transaction traceability, factory reliability and market planning around shared records and permissions. It is not a claim that every component is new or that all modules are already implemented. Existing systems will be assessed and integration preferred where it offers a better path than replacement.

6. Conclusion

SukariSol is a proposed integrated digital platform intended to connect decisions across Kenya's sugar value chain. Its six modules address farm productivity intelligence, cane logistics, payment reconciliation, factory maintenance, by-product commercialization and market planning. The practical route forward is to validate the most costly problems, secure a pilot partner, test data availability and demonstrate one end-to-end workflow before scaling. By linking reliable records to specific operational decisions, SukariSol aims to provide a credible pathway toward improved coordination, transparency, efficiency and sustainable value creation in the sugar industry.

References

[1] Kenya Sugar Board. 2026 Innovation Call.

[2] National agriculture sector reporting on sugar-sector operational constraints (full citation to be inserted by the authors; not included in the source draft).